

## LEIPZIG HOLZHAUSEN, Germany

WMO: 104710

Lat: 51.3150N

Lon: 12.4461E

Elev: 138

StdP: 99.68

Time Zone: 1.00 (EUC)

Period: 97-13

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -11.0      | -8.2 | -13.8                         | 1.2 | -9.8 | -11.1 | 1.5 | -6.8 | 8.2                   | 8.9  | 7.3 | 7.2  | 2.1                   | 80   | 0.477 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.6                    | 30.3            | 19.3 | 28.3 | 18.4 | 26.5 | 17.9 | 20.5                | 27.3 | 19.6 | 25.7 | 18.8 | 24.4 | 2.3                  | 110  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 18.3                            | 13.4 | 22.2 | 17.5 | 12.7 | 21.6 | 16.7 | 12.1 | 21.0 | 59.3          | 27.3 | 56.4 | 25.8 | 53.8 | 24.5 | 24.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 6.5           | 5.7  | 5.0 | DB                         | -13.6 | 34.3               | 3.7 | 1.9                                                | -16.3 | 35.7       | -18.5 | 36.8       | -20.6 | 37.9       | -23.3 | 39.3 | (4) |
| (5)               |      |     | WB                         | -14.0 | 22.2               | 3.7 | 0.8                                                | -16.7 | 22.8       | -18.8 | 23.3       | -20.9 | 23.8       | -23.5 | 24.4 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 10.1   | 1.2   | 1.8   | 5.1   | 10.1  | 14.4  | 17.2  | 19.2  | 19.1  | 15.0  | 10.1  | 5.4   | 1.7   |  |  |
|                                                                        | DBStd       | 7.71   | 5.19  | 4.98  | 4.26  | 4.00  | 3.68  | 3.50  | 3.17  | 3.08  | 3.30  | 3.80  | 3.66  | 4.31  |  |  |
|                                                                        | HDD10.0     | 1168   | 274   | 232   | 159   | 48    | 6     | 1     | 0     | 0     | 2     | 47    | 142   | 258   |  |  |
|                                                                        | HDD18.3     | 3169   | 532   | 463   | 412   | 249   | 131   | 62    | 26    | 27    | 108   | 257   | 387   | 516   |  |  |
|                                                                        | CDD10.0     | 1193   | 1     | 2     | 7     | 49    | 143   | 217   | 287   | 282   | 152   | 49    | 4     | 1     |  |  |
|                                                                        | CDD18.3     | 153    | 0     | 0     | 0     | 1     | 9     | 28    | 54    | 51    | 9     | 1     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 1327   | 0     | 0     | 0     | 12    | 101   | 248   | 492   | 398   | 75    | 1     | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 375    | 0     | 0     | 0     | 2     | 19    | 65    | 166   | 114   | 9     | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 2.3    | 2.7   | 2.7   | 2.7   | 2.3   | 2.3   | 2.1   | 2.0   | 2.0   | 2.0   | 2.2   | 2.3   | 2.7   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 515    | 33    | 27    | 35    | 38    | 50    | 56    | 61    | 62    | 45    | 36    | 39    | 37    |  |  |
|                                                                        | PrecMax     | 756    | 79    | 82    | 97    | 114   | 124   | 113   | 155   | 158   | 139   | 131   | 102   | 84    |  |  |
|                                                                        | PrecMin     | 301    | 1     | 2     | 5     | 3     | 5     | 11    | 7     | 13    | 6     | 3     | 0     | 1     |  |  |
|                                                                        | PrecStd     | 108    | 17    | 16    | 21    | 23    | 28    | 24    | 32    | 32    | 25    | 26    | 21    | 17    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 13.1  | 14.4  | 18.6  | 24.6  | 29.0  | 31.4  | 33.7  | 32.8  | 27.9  | 22.1  | 15.6  | 12.8  |  |  |
|                                                                        |             | MCWB   | 9.5   | 9.9   | 11.5  | 15.1  | 17.7  | 20.5  | 20.3  | 19.3  | 18.0  | 16.4  | 12.0  | 9.6   |  |  |
|                                                                        | 2%          | DB     | 11.1  | 12.0  | 15.9  | 22.0  | 26.3  | 28.6  | 30.9  | 29.5  | 25.5  | 19.1  | 13.4  | 10.4  |  |  |
|                                                                        |             | MCWB   | 8.4   | 8.9   | 10.5  | 13.7  | 17.1  | 18.9  | 19.1  | 19.1  | 17.1  | 14.5  | 10.4  | 7.9   |  |  |
|                                                                        | 5%          | DB     | 9.6   | 10.3  | 13.4  | 19.7  | 23.7  | 26.2  | 28.3  | 27.5  | 23.3  | 17.3  | 11.7  | 8.8   |  |  |
|                                                                        |             | MCWB   | 7.3   | 7.5   | 9.0   | 12.6  | 15.7  | 17.9  | 18.6  | 18.2  | 16.3  | 13.6  | 9.4   | 6.9   |  |  |
|                                                                        | 10%         | DB     | 7.8   | 8.7   | 11.4  | 17.4  | 21.5  | 24.1  | 26.0  | 25.4  | 21.1  | 15.6  | 10.5  | 7.5   |  |  |
|                                                                        |             | MCWB   | 5.9   | 6.5   | 8.0   | 11.4  | 14.4  | 17.0  | 18.0  | 17.7  | 15.6  | 12.7  | 8.5   | 5.8   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 10.2  | 10.2  | 12.2  | 16.1  | 19.2  | 22.1  | 21.4  | 21.3  | 19.2  | 16.6  | 12.4  | 10.0  |  |  |
|                                                                        |             | MCDB   | 12.5  | 13.3  | 16.8  | 23.9  | 26.8  | 28.0  | 29.7  | 29.3  | 25.3  | 20.5  | 15.4  | 12.0  |  |  |
|                                                                        | 2%          | WB     | 8.8   | 9.3   | 10.7  | 14.4  | 17.8  | 20.0  | 20.3  | 20.3  | 17.9  | 15.2  | 10.7  | 8.1   |  |  |
|                                                                        |             | MCDB   | 10.6  | 11.8  | 14.6  | 20.3  | 24.1  | 26.5  | 27.1  | 27.0  | 23.2  | 18.4  | 12.7  | 9.9   |  |  |
|                                                                        | 5%          | WB     | 7.4   | 7.9   | 9.6   | 13.2  | 16.6  | 18.8  | 19.5  | 19.4  | 17.0  | 13.9  | 9.8   | 7.0   |  |  |
|                                                                        |             | MCDB   | 9.3   | 9.9   | 13.1  | 18.5  | 22.1  | 24.5  | 25.6  | 24.9  | 21.6  | 16.6  | 11.5  | 8.8   |  |  |
|                                                                        | 10%         | WB     | 6.0   | 6.4   | 8.3   | 12.0  | 15.4  | 17.6  | 18.7  | 18.5  | 16.1  | 12.9  | 8.7   | 5.8   |  |  |
|                                                                        |             | MCDB   | 7.7   | 8.7   | 11.1  | 16.6  | 20.3  | 22.6  | 24.5  | 23.4  | 20.5  | 15.3  | 10.3  | 7.3   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 4.8   | 5.6   | 7.2   | 9.5   | 9.9   | 9.5   | 9.6   | 9.7   | 8.9   | 7.2   | 5.1   | 4.3   |  |  |
|                                                                        |             | MCDBR  | 5.8   | 7.2   | 11.1  | 13.7  | 13.6  | 13.3  | 14.2  | 13.3  | 12.9  | 9.6   | 6.6   | 5.5   |  |  |
|                                                                        | 5% WB       | MCWBR  | 4.4   | 5.0   | 6.5   | 6.9   | 6.1   | 5.4   | 4.8   | 4.9   | 5.8   | 5.7   | 4.5   | 4.2   |  |  |
|                                                                        |             | MCWBR  | 5.5   | 6.1   | 9.7   | 12.4  | 12.0  | 11.9  | 11.0  | 10.8  | 8.9   | 6.1   | 5.5   | 5.5   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.316 | 0.340 | 0.384 | 0.418 | 0.420 | 0.416 | 0.434 | 0.427 | 0.389 | 0.366 | 0.339 | 0.310 |  |  |
|                                                                        | taud        |        | 2.364 | 2.330 | 2.227 | 2.181 | 2.212 | 2.252 | 2.210 | 2.248 | 2.334 | 2.373 | 2.388 | 2.366 |  |  |
|                                                                        | Ebn at Noon |        | 696   | 776   | 802   | 818   | 835   | 840   | 816   | 801   | 792   | 736   | 659   | 638   |  |  |
|                                                                        | Edh at Noon |        | 63    | 85    | 113   | 132   | 134   | 130   | 134   | 123   | 101   | 81    | 62    | 54    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 0.74  | 1.45  | 2.45  | 3.99  | 4.87  | 5.21  | 4.93  | 4.27  | 3.07  | 1.79  | 0.85  | 0.57  |  |  |
|                                                                        | RadStd      |        | 0.10  | 0.24  | 0.32  | 0.51  | 0.58  | 0.43  | 0.61  | 0.38  | 0.39  | 0.21  | 0.12  | 0.06  |  |  |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) | Regional (52 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | +0.33 | N/A         | N/A     | N/A     | +22     |

Nomenclature: See separate page